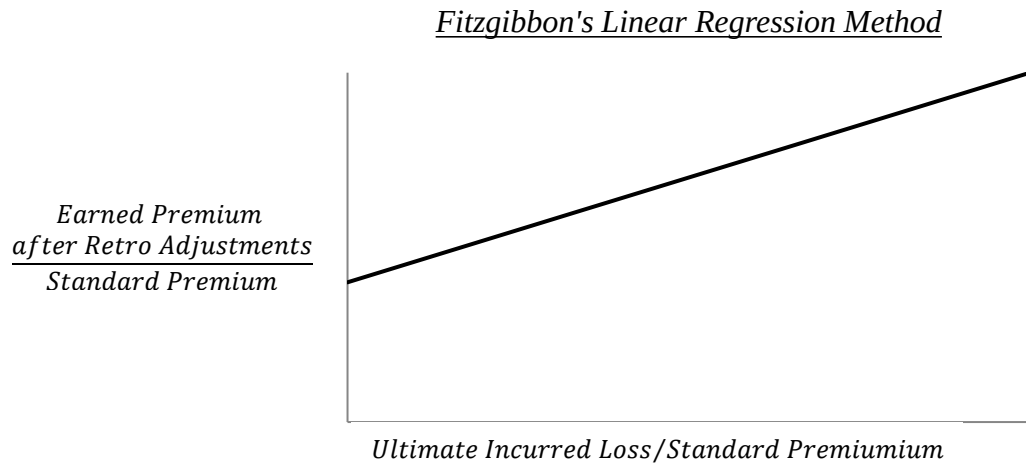


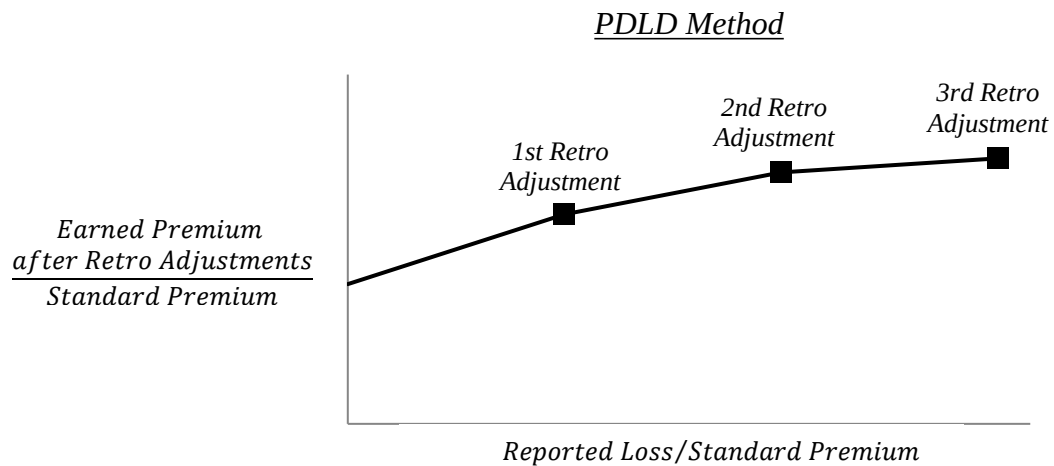
18. (2 points)

In evaluating the premium asset for a property and casualty insurance company's book of retrospectively rated policies as of December 31, 2016, the following methods have been considered:

- Chain ladder development method on historical triangles of earned premium using Part 6 of Schedule P.
- Fitzgibbon's Linear Regression method, which produced the following graph:



- Premium Development to Loss Development (PDL) method using the retro formula and average retro parameters, which produced the following graph:



CONTINUED ON NEXT PAGE

(18 continued)

a. (0.5 point)

Briefly describe one advantage and one disadvantage of using the chain ladder development method on historical earned premium triangles.

b. (0.25 point)

Briefly describe one disadvantage of using Fitzgibbon's linear regression method.

c. (0.25 point)

Explain why the slope factor in Fitzgibbon's linear regression method is typically not exactly unity.

d. (0.5 point)

Briefly describe one advantage and one disadvantage of using the PDLD method.

e. (0.5 point)

Explain why the slopes differ between Fitzgibbon's linear regression method and the PDLD method.

CONTINUED ON NEXT PAGE

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 18	
TOTAL POINT VALUE: 2	LEARNING OBJECTIVE(S): A14a
SAMPLE ANSWERS	
Part a: 0.5 point	
<u>Sample Responses for Advantages include any of the following:</u> • Direct method of developing prem as focuses on premium itself • Responsive to how premiums have developed historically • As Feldbloom argues, it is more consistent with the way we develop losses • Easy to calculate and understand <u>Sample Responses for Disadvantages include any of the following:</u> • Premium data/adjustments are often delayed around 9 months and since premium depends on losses -> quicker results (less lag) if use losses instead • Will not pick up any parameter changes that go into the premium such as changes in expenses that impact the basic premium or changes in the tax multiplier • May be quite volatile especially if the business mix has changed and does not reflect the retrospective rating logic • If changing contract parameters, the chain latter method will no longer be representative of the expected development	
Part b: 0.25 point	
<u>Sample Responses include any of the following:</u> • Assumes constant prem-to-loss development -> not accurate since more and more losses are capped in later adjustments • Fitzgibbon's method is only based on ultimate losses so if your reporting pattern isn't coming in as expected there is no way to get back on track to estimate future retro premiums accurately • It calculates a fixed amount of ultimate premium without consideration of the amount of premium actually emerged and booked at last period. So it created a potential large difference and no way to fix it.	
Part c: 0.25 point	
<u>Sample Responses include any of the following:</u> • The "y-intercept will be >0 due to fixed expenses, profit, etc to cover -> then for each dollar of loss, some will be capped or hit min/max in plan -> if slope was 1.0 insured would be paying back for all losses + expenses -> wouldn't make sense • There are min/max premiums and loss capping in retro rating plans • Part of losses above maximum and per occurrence limit - Minimum above Basic premium in certain plans - LCF and tax multiplier effect	
Part d: 0.5 point	
<u>Sample Responses for Advantages include any of the following:</u> • Losses develop more quickly than premiums -> get quicker results/estimates • Based directly on the retro rating formula • Timeliness: can calculate premium asset as soon as losses are reported and booked and	

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

update each quarter • It is more responsive to the current parameters sold. <u>Sample Responses for Disadvantages include any of the following:</u> • Parameters to use in formulas may be difficult to estimate if book of business has been changing • Can be difficult to estimate the parameters such as the basic premium and tax multipliers that go into the retro rating formula • There could be a bias due to application of average parameters in the PDLD formula
Part e: 0.5 point <u>Sample Responses include any of the following:</u> • PDLD slopes start steep and then become more and more shallow -> recognizing more and more losses capped in later adjustments -> PDLD -> this is an improved (more accurate process than Fitzgibbons • The PDLD method recognizes that premium responsiveness to losses decreases with each retro adjustment as more losses become capped and the min/max premiums get applied, while Fitzgibbon's method is only based on premium responsiveness to ultimate losses, not at each retro adjustment • PDLD slope are decreasing as more losses are capped by maximum limits and per occurrence limits as time passes. Fitzgibbon does not take this effect in consideration so slope is constant
EXAMINER'S REPORT Candidates were expected to have general knowledge of premium development, and the methods discussed in Teng/Perkins/Feldblum to address this development. Specific methods were listed with illustrations to make it clear precisely which method was to be discussed. In general, candidates struggled on less commonly used methods.
Part a Candidates were expected to demonstrate they knew an advantage and disadvantage of the chain ladder method which were specific to premium development. Common errors include: • Listing general advantages/disadvantages of the chain ladder method that were not specifically tied to the issues involved in retrospective policies (change in retro parameters). • Stating that this method used losses. • Stating that the method used data/experience, but not stating what data/experience was used. • For disadvantages, stating that not using losses was a disadvantage without linking to the lag between premium and losses.
Part b Candidates were expected to demonstrate they knew a disadvantage of the Fitzgibbon method. Common errors include: • Stating that a disadvantage of the method was that it relied on developing ultimate

EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

losses. • Stating that the method did not react to a deviation from expected losses, which is not true. The method reacts to loss deviations through the ultimate loss selection. It does not react to a deviation from expected premium, or the ratio between premium and losses. • Listing an advantage instead of a disadvantage.
Part c Candidates were expected to demonstrate they knew any reason that directly caused the slope of Fitzgibbon to not be equal to 1. Common errors include: • Not understanding the meaning of unity. • Restating the question without further explanation.
Part d Candidates were expected to demonstrate they knew an advantage and disadvantage of the formula-based PDLDM method with the Feldblum adjustment. Common errors include: • Responding with specific advantages of the PDLDM without Feldblum's adjustment. • Giving an advantage/disadvantage related to the empirical PDLDM. • Only stating that "it is more accurate" or "it is more difficult".
Part e Candidates were expected to demonstrate they understood why the PDLDM would have different slopes than Fitzgibbon, and to identify a cause. Several of each are listed in both the Teng & Perkins reading and the Feldblum reading. Common errors include: • Stating that the different slopes were caused by Fitzgibbon fitting a regression on Ultimate loss, while the PDLDM fit on incurred loss.